Many Alternatives, Random Incentives: Sharp Bounds and Efficient Estimation in Treatment Selection Models

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October 30, 2026

Motivation: Policy Evaluation with Endogenous Treatments

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One solution: use **exogenous variation** that shifts choices, but not potential outcomes.

One Possible Solution: Instrumental Variables

Binary treatment: under a "monotonicity" assumption, binary IV estimand identifies LATE (Imbens and Angrist, 1994)

- in some cases, LATE is policy relevant
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Multiple treatments: point identification can be hard in practice.
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- no longer sufficient for point-identification of relevant parameters

$\rightarrow$ people can respond differently, and across different treatment margins

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What if we have **multiple instruments?** More identifying power, but

- not always obvious what the "natural" assumptions are
- ideally: incorporate beliefs or knowledge about the instrument(s)

Multiple IVs: The Problem with "Monotonicity"

Consider two ways of encouraging college attendance:

- Z_1: Tuition Subsidy
- Z_2: Transport Assistance

Z_1 and Z_2 satisfy monotonicity separately.
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But, with heterogenous agents:

Person	Subsidy only	Transport only
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Goal: use instruments we have, to learn about parameters we care about, under assumptions we believe (or can choose flexibly)

This Paper: Flexible and Sharp Partial Identification Approach

Setting:

- "Instrument" $Z \in \mathcal{Z}$: randomly assigned, categorical
- "Treatment" $D \in \mathcal{D}$: generally endogenous, categorical
- Outcome(s) $Y \in \mathcal{Y} \subseteq \mathbb{R}$: endogenous, may be continuous

$\rightarrow$ D and Z may be ordered or unordered, may be multi-dimensional

Scope: Models that restrict how individuals select into treatment in response to instruments/incentives/assignment

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- **Moment inequalities** characterization, sharp for "full" independence
- Enables **estimation and inference** without discretization

  $\rightarrow$ Substantial gains in statistical power and tractability when Y continuous

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Randomly assigned instrument $Z = (Z_T, Z_V)$, where

- $Z_T \in \{0, 1\}$: Incentivizes training participation
- $Z_V \in \{0, 1\}$: Encourages actions that make the training more salient

Credentials Example: Modelling Selection

Impose the following restrictions on selection of (T, V):

1. $Z_T : 0 \to 1$ makes all individuals weakly more likely to get training
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Define potential treatments $D(Z) := (T(Z_T), V(T, Z_V))$

Let Y be income, and define potential outcome $Y(d) = Y(t, v)$ as the income earned if D fixed at $(T, V) = (t, v)$.

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Let Y be income, and define potential outcome $Y(d) = Y(t, v)$ as the income earned if D fixed at $(T, V) = (t, v)$. Can decompose effects:

$$\underbrace{Y(1,1) - Y(0,0)}_{\text{Total Effect}} = \underbrace{Y(1,1) - Y(1,0)}_{\text{Additional Signalling Effect}} + \underbrace{Y(1,0) - Y(0,0)}_{\text{Baseline Training Effect}}$$

Credentials Example: Intent-to-Treat

Let $\mathcal{C} := \{T(1) = 1, T(0) = 0\}$ be the "training compliers".

For each $z_V \in \{0, 1\}$, define the "intent-to-treat" parameter as

$$\begin{aligned}\mathrm{ITT}_{Z_T}(z_v) &:= \mathbb{E}[Y \mid Z_T = 1, Z_V = z_v] - \mathbb{E}[Y \mid Z_T = 0, Z_V = z_v] \\ &= \mathrm{P}[\mathcal{C}, V(1, z_v) = 1] \underbrace{\mathbb{E}[Y(1,1) - Y(0,0) \mid \mathcal{C}, V(1, z_v) = 1]}_{\text{Total Effect}} \\ &\quad + \mathrm{P}[\mathcal{C}, V(1, z_v) = 0] \underbrace{\mathbb{E}[Y(1,0) - Y(0,0) \mid \mathcal{C}, V(1, z_v) = 0]}_{\text{Baseline Training Effect}}\end{aligned}$$

For any subpopulation G, can also consider parameters like:

$$\begin{aligned}\mathrm{Training}_G &:= \mathbb{E}[Y(1,0) - Y(0,0) \mid G] \ , \\ \mathrm{SignallingShare}_G &:= \frac{\mathbb{E}[Y(1,1) - Y(1,0) \mid G]}{\mathbb{E}[Y(1,1) - Y(0,0) \mid G]} =: \frac{\mathrm{Signalling}_G}{\mathrm{Total}_G} \ .\end{aligned}$$

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Credentials Example: 2 DGPs that agree on all ITTs

Consider 2 Populations A and B, with :

Response pair	$(T(0), T(1))$	$(W(0), W(1))$	Marginal Probability
$(0,0)$	Never-taker N_T	Never-visible N	0.20
$(0,1)$	Training complier C	Visibility responder R	0.65
$(1,1)$	Always-taker A_T	Always-visible H	0.15

where $W(z_V) = V(1, z_V)$ describes visibility if trained.

For compliers, generate outcomes as gaussians with the means:

Population	Visibility type	$Y(0,0)$ mean	$Y(1,0)$ mean	$Y(1,1)$ mean
A	N or R	48	56	60
A	H	48	48.5	52
B	N or R	48	50	54
B	H	48	48.5	86

For T-never-takers, subtract 2 from each mean; T-always-takers, add 2.

Credentials Example: 2 DGPs that agree on all ITTs

	Contrast	Y ($1,000s)
Both populations satisfy	$\text{ITT}_{Z_T}(0)$	4.810
	$\text{ITT}_{Z_T}(1)$	6.500
	$\text{ITT}_{Z_V}(0)$	0.390
	$\text{ITT}_{Z_V}(1)$	2.080

But, for $G := \{T(0) = 0,\ T(1) = 1\} \cap \{V(1,0) = 0,\ V(1,1) = 1\}$, we have:

Controlled target within G_R	Population A	Population B
$\mathbb{E}[Y(1,0) - Y(0,0) \mid G_R]$	$[6.355,\ 9.645]$	$[0.355,\ 3.645]$
$\mathbb{E}[Y(1,1) - Y(1,0) \mid G_R]$	4 (point identified)	4 (point identified)
$\mathbb{E}[Y(1,1) - Y(0,0) \mid G_R]$	$[10.355,\ 13.645]$	$[4.355,\ 7.645]$

Structural IV Model with Essential Heterogeneity

Introduce unobservables $(\boldsymbol{V}, \varepsilon_Y)$, functions g_D and g_Y:

Treatment Selection Equation: $D = g_D(Z, \boldsymbol{V})$,

Outcome Equation: $Y = g_Y(D, \boldsymbol{V}, \varepsilon_Y)$,

Independence Condition: $\underbrace{Z, \boldsymbol{V}, \varepsilon_Y \text{ are mutually independent.}}_{\text{Equivalent to Random Assignment of } Z}$

Formally equivalent to the PO model, can define

$$Y_d := g_Y(d, \boldsymbol{V}, \varepsilon_Y) \quad \text{and} \quad D_z := g_D(z, \boldsymbol{V}) \ .$$

Define vector of counterfactual choices ("principal strata"):

$$(D_z : z \in \mathcal{Z}) = (g_D(z, \boldsymbol{V}) : z \in \mathcal{Z})$$

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- Example (Vytlacil 2002): $D = \mathbb{1}\{\boldsymbol{V} \leq \nu(Z)\}$, $\boldsymbol{V} \sim \mathrm{Unif}[0,1]$, $\nu \nearrow$

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$\rightarrow$ Such models are characterized by restrictions on *distribution* of $\boldsymbol{T}$

Other Examples of TSMs and Applications

Monotonicity: Imbens & Angrist (1996), Rose & Shem-Tov (2021).

Revealed Preference: Kline & Walters (2016), Kirkebøen, Leuven, & Mogstad (2016), Heckman & Pinto (2018), Pinto (2021).

Encouragement Designs, Generalized Threshold Crossing Models: Heckman and Vytlacil (2001, 2005), Lee and Salanié (2018), Bai, Huang, Tabord-Meehan (2024)

Many IVs: Mogstad, Torgovitsky, & Walters (2020), Mountjoy (2022), Goff (2024), Sun and Wüthrich (2024), Sigstad (2024), van't Hoff et al. (2025).

Mediation and Mechanisms: Imai, Keele, & Yamamoto (2010), Frolich & Huber (2017), Kwon & Roth (2026).

Sample Selection and Non-Response Bias: Lee (2009), Dutz et al. (2026), Kroft, Mourifié, & Vayalinkal (2026).

Overview of Results: Identification

1. Sharp testable implications of such models as **moment inequalities**

$\rightarrow$ Finite-dimensional convex representation (even if Y continuous)

$\rightarrow$ **Sharp Specification Test** for TSMs & IV Validity

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2. Partial identification of $\mathbb{E}\left[Y_{d_1} - Y_{d_2} \,\middle|\, \boldsymbol{V} \in \boldsymbol{\mathcal{V}}^*\right]$ and $\frac{\mathbb{E}\left[Y_{d_1} - Y_{d_2} \,\middle|\, \boldsymbol{V} \in \boldsymbol{\mathcal{V}}^*\right]}{\mathbb{E}\left[Y_{d_3} - Y_{d_4} \,\middle|\, \boldsymbol{V} \in \boldsymbol{\mathcal{V}}^*\right]}$

$\rightarrow$ **Sharp Bounds** and Parameter-inclusion Tests

These results enable inference via Chernozhukov, Lee and Rosen (2013).

Overview of Results: Estimation

Estimation of conditional means and treatment effects:

1. **Semiparametric efficiency bound** for estimation of sharp bounds
2. Formulate tractable "plug-in" estimator for sharp bounds
   $\rightarrow$ Computation via finite-dimensional convex programs
3. Propose new cross-fitted bias-corrected estimator for the bounds
   $\rightarrow$ Attains efficiency bound in suitably regular cases
   $\rightarrow$ Mitigates inward bias of "plug-in" estimator in irregular cases

Related IV Identification Literature

Point Identification: Angrist and Imbens (1995)... Kirkebøen et al. (2016); Heckman and Pinto (2018); Lee and Salanié (2018); Mountjoy (2022)...

Partial Identification: Manski (1989, 1990); Balke and Pearl (1997)... Beresteanu et al. (2012)... Mogstad et al. (2018); Kitagawa (2021); Marx (2024)... Bai, Huang, Moon, Santos, Shaikh, and Vytlacil (2026)

→ **This paper:** sharp bounds under full IV independence for selection models with multivalued instruments and continuous outcomes

Specification Testing : Kitagawa (2015); Mourifié and Wan (2017); Kédagni and Mourifié (2020); Sun (2023); Bai, Huang, and Tabord-Meehan (2025); Kaido and Ponomarev (2025)...

→ **This paper:** sharp test for general selection models with categorical IV

"Trimming" Bounds of Horowitz and Manski (1995)

Suppose all we observe is the following mixture distribution, with CDF F,

$$F(y) = \gamma F_1(y) + (1-\gamma) F_2(y) \ ,$$

where $\gamma \in (0,1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

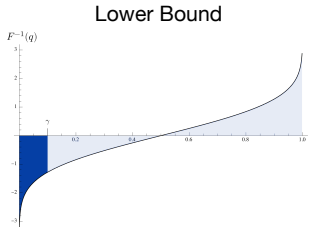
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where $\gamma \in (0,1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

If $\gamma = 0.1$, sharp lower bound for $E[Y_1]$ is constructed as follows:



(a) Lower bound is area of shaded region

Figure 1: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.1$

“Trimming” Bounds of Horowitz and Manski (1995)

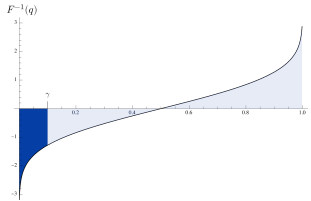
Suppose all we observe is the following mixture distribution, with CDF F,

$$F(y) = \gamma F_1(y) + (1-\gamma) F_2(y) \ ,$$

where $\gamma \in (0,1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

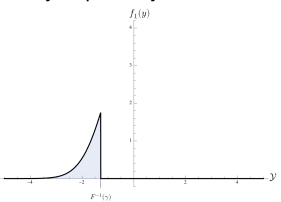
If $\gamma = 0.1$, sharp lower bound for $E[Y_1]$ is constructed as follows:

Lower Bound



(a) Lower bound is area of shaded region

Density Implied by Lower Bound



(b) Density of F_1

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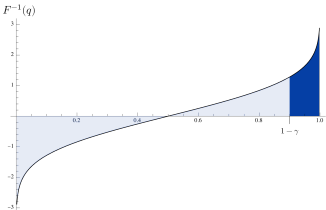
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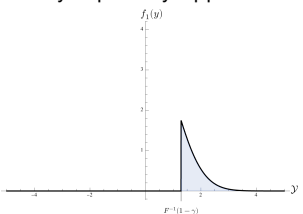
If $\gamma = 0.1$, sharp **upper** bound for $E[Y_1]$ is constructed as follows:

Upper Bound



(a) Upper bound is area of shaded region

Density Implied by Upper Bound



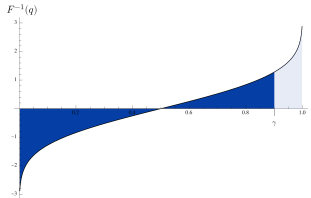
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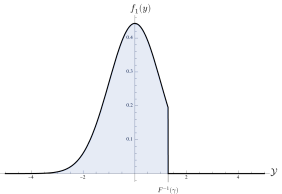
If instead we have $\gamma = 0.9$, then lower bound is

Lower Bound



(a) Lower bound is area of shaded region

Density Implied by Lower Bound

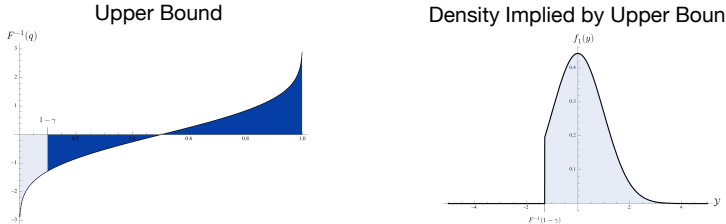


(b) Density of F_1

Figure 2: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.9$

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Horowitz and Manski (1995) Bounds: Takeaway

The observed mixture distribution, written in terms of densities, is

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- $\rightarrow$ optimizing the right hand side over all functions $\gamma f_1(\cdot)$ that satisfy the two conditions above results in the Horowitz-Manski bounds

Treatment Selection Models and "Response-Types"

A ***treatment selection model*** (TSM) restricts the selection mechanism

$$D = g_D(Z, \boldsymbol{V}) \ .$$

$\rightarrow$ A TSM $\mathbb{S}$ is specified as a collection of admissible pairs $(F_{\boldsymbol{V}}, g_D)$

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$\rightarrow$ In fact, $\mathbf{S}$ has identifying power *only* through restrictions on $\boldsymbol{\pi}$

Identification Power of TSMs

Denote by Q_0 the (unknown) distribution of $((Y_d : d \in \mathcal{D}), \boldsymbol{T}, Z)$.

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Define $\mathcal{R}_{\boldsymbol{\pi}} = \mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S})$ as the set of *a priori* restrictions on distribution of $\boldsymbol{T}$,

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$$\begin{aligned} \mathcal{R}_{\boldsymbol{\pi}} &:= \left\{ \text{probability vector } \boldsymbol{\pi} \;\middle|\; \boldsymbol{\pi}_{(\text{defier})} = 0 \right\} \\ &= \left\{ \boldsymbol{\pi} \text{ is a probability vector on } \boldsymbol{\mathcal{T}}_{\mathbf{S}} \right\}. \end{aligned}$$

$\rightarrow$ $\mathcal{R}_{\boldsymbol{\pi}}$ is finite dimensional, always exists

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Theorem 1.

For any treatment selection model $\mathbf{S}$, $\mathcal{Q}_{\mathbf{S}} = \left\{ Q \in \mathcal{Q} \,\middle|\, \boldsymbol{\pi}_{Q} \in \mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S}) \right\}$.

IV Mixture Model: Basic Idea

For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

We observe Y for those assigned $Z = z$ and took treatment d

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Identification under TSMs: Pointwise Restrictions

Can write these mixtures in terms of densities, with coefficient matrix $\mathbf{R}_d$:

$$\mathbf{R}_d \boldsymbol{q}_d(y) = \boldsymbol{p}_d(y) \quad \text{and} \quad \boldsymbol{q}_d(y) \geq 0$$

$\rightarrow$ ${\boldsymbol{p}_d}^P$: observed densities $\boldsymbol{p}_{d|Z=z} = \mathbb{P}_P(D = d|Z = z) f_{Y|D=d,Z=z}$

$\rightarrow$ ${\boldsymbol{q}_d}^{Q_0}$ unobserved type-conditional densities $\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\, f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}}$

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, unrestricted selection)

In this case, the response types are always-takers (AT), compliers (C), defiers (D), never-takers (NT) , leading to

$$\mathbf{R}_0 = \begin{array}{c} \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \begin{matrix} (Z=0) \\ (Z=1) \end{matrix} \quad \text{and} \quad \mathbf{R}_1 = \begin{array}{c} \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Identification under TSMs: Mass Restrictions

Pointwise restrictions on $\boldsymbol{q}_d$ $\forall y$: $\mathbf{R}_d \boldsymbol{q}_d(y) = \boldsymbol{p}_d(y)$ and $\boldsymbol{q}_d(y) \geq 0$.

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But $\boldsymbol{q}_d$ is a candidate for $\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau}) f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}}$, so must also satisfy

$$\int_{\mathcal{Y}} \boldsymbol{q}_d(y)\, dy = \left(\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\right)_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}} = \boldsymbol{\pi}^{Q_0} \ .$$

$\boldsymbol{\pi}^{Q_0}$ doesn't depend on d $\rightarrow$ this must hold for every d, with **same** $\boldsymbol{\pi}$.

$\rightarrow$ That $\boldsymbol{\pi}$ must also satisfy $\boldsymbol{\pi} \in \mathcal{R}_{\boldsymbol{\pi}}$.

Identification under TSMs: Mass Restrictions

Pointwise restrictions on $\boldsymbol{q}_d$ $\forall y$: $\quad \mathbf{R}_d \boldsymbol{q}_d(y) = \boldsymbol{p}_d(y)$ and $\boldsymbol{q}_d(y) \geq 0$.

$\rightarrow$ Has a solution $\iff \boldsymbol{p}_d^P(y) \in \text{cone}(\mathbf{R}_d) \ \ \forall y \in \mathcal{Y}$

But $\boldsymbol{q}_d$ is a candidate for $\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau}) f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}}$, so must also satisfy

$$\int_{\mathcal{Y}} \boldsymbol{q}_d(y)\, dy = \left(\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\right)_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}} = \boldsymbol{\pi}^{Q_0} \ .$$

$\boldsymbol{\pi}^{Q_0}$ doesn't depend on d $\rightarrow$ this must hold for every d, with **same** $\boldsymbol{\pi}$.

$\rightarrow$ That $\boldsymbol{\pi}$ must also satisfy $\boldsymbol{\pi} \in \mathcal{R}_{\boldsymbol{\pi}}$.

$\rightarrow$ Identified set for parameters θ is set of all values for θ that can be generated by $(\boldsymbol{q}_d)_{d \in \mathcal{D}}$ satisfying these two restrictions

Identification under TSMs: Overview of Approach

1. Actually, we can work with the relaxed system:

$$\mathbf{R}_d \boldsymbol{q}_d(y) \leq \boldsymbol{p}_d(y) \quad \text{and} \quad \boldsymbol{q}_d(y) \geq 0 \qquad (\star)$$

Lossless once we require that

1. $\int_{\mathcal{Y}} \boldsymbol{q}_d(y)\, dy \in \mathcal{R}_{\boldsymbol{\pi}}$
2. $\int_{\mathcal{Y}} \boldsymbol{q}_d(y)\, dy$ is invariant to d

Key: $(\star)$ always has a solution!

2. Each solution $\boldsymbol{q}_d$ of $(\star)$ integrates to a "candidate" value of

$$\boldsymbol{\pi} := \left(\mathbb{P}\left(\boldsymbol{T} = \boldsymbol{\tau}\right)\right)_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}}$$

For each d, call Π_d the set of such candidate values.

$$\rightarrow \mathbf{S} \text{ rejected } \iff \mathcal{Q}_{\mathbf{S}} = \emptyset \iff \mathcal{R}_{\boldsymbol{\pi}} \cap \bigcap_{d \in \mathcal{D}} \Pi_d = \emptyset$$

$q_d(y)$ must be a solution to $(\star)$

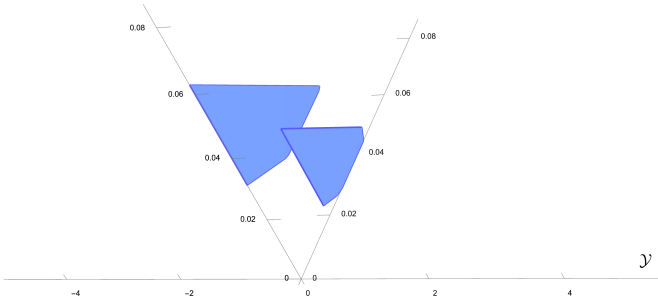


Figure 3: Hypothetical graph of the correspondence $y \mapsto \{\boldsymbol{q}_d \geq 0 \,|\, \text{ solves } (\star)\}$

$\boldsymbol{q}_d(y)$ must be a solution to $(\star)$

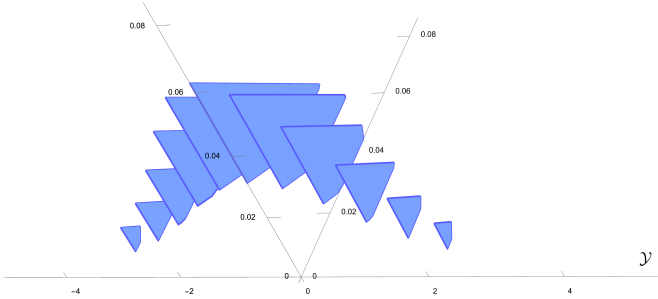


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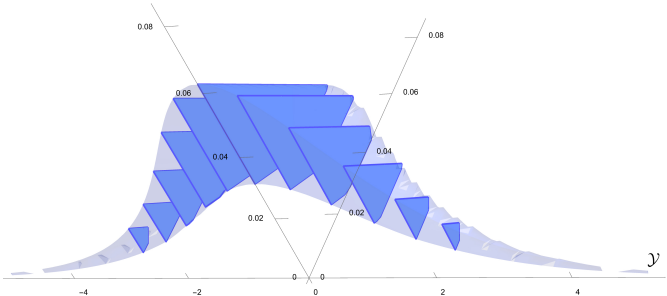


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$q_d(y)$ must be a solution to $(\star)$

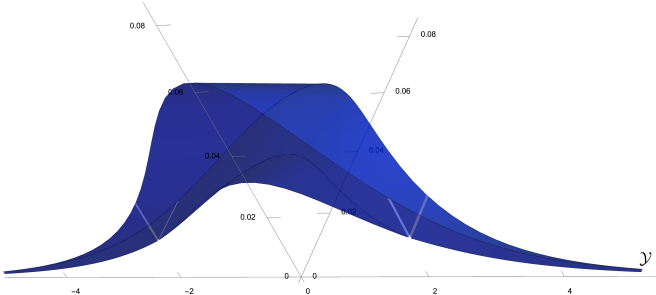


Figure 3: Hypothetical graph of the correspondence $y \mapsto \{\boldsymbol{q}_d \geq 0 \,|\, \text{ solves } (\star)\}$

$q_d(y)$ must be a solution to $(\star)$

$\rightarrow$ If $\mathbf{x}(y) \in \mathcal{B}_d(p_d(y)) := \{q_d \geq 0 \mid \text{ solves } (\star)\}$ for all $y \in \mathcal{Y}$, then $\mathbf{x}$ is a "candidate" for $q_d(y)$.

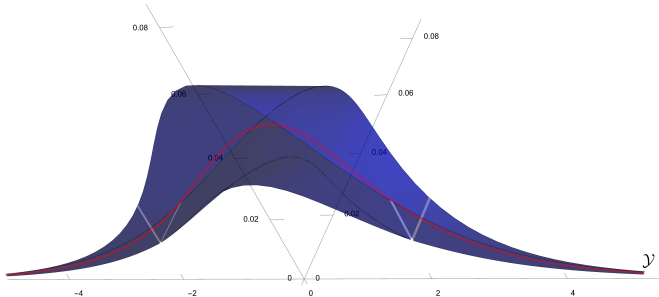


Figure 3: Hypothetical graph of the correspondence $y \mapsto \{q_d \geq 0 \mid \text{ solves } (\star)\}$

IV Mixture Model: Informal Overview of Approach

2. Each solution $\boldsymbol{q}_d$ of $(\star)$ integrates to a "candidate" value of

$$\boldsymbol{\pi} := \left(\mathbb{P}\left(\boldsymbol{T} = \boldsymbol{\tau}\right)\right)_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}}$$

For each d, call Π_d the set of such candidate values.

$\rightarrow$ S rejected $\iff \mathcal{Q}_{\mathbf{S}} = \emptyset \iff \mathcal{R}_{\boldsymbol{\pi}} \cap \bigcap\limits_{d \in \mathcal{D}} \Pi_d = \emptyset$

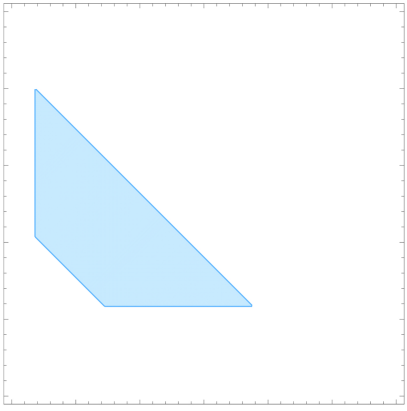
Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, unrestricted selection)

Since $\mathbf{R}_1 = \begin{array}{c} \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$ and $\boldsymbol{q}_d \geq 0$,

$$\mathbf{R}_1 \boldsymbol{q}_d(y) = \boldsymbol{p}_d(y) \implies \boldsymbol{q}_{d|\boldsymbol{T}=\text{AT}}(y) \leq \min\left\{\boldsymbol{p}_{1|Z=0}(y), \boldsymbol{p}_{1|Z=1}(y)\right\}$$

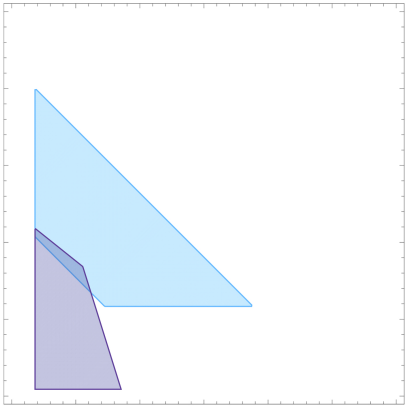
$$\implies \boldsymbol{\pi}_{(\text{AT})} \leq \int_{\mathcal{Y}} \min\left\{\boldsymbol{p}_{1|Z=0}(y), \boldsymbol{p}_{1|Z=1}(y)\right\} dy$$

(Stylized) Example with 3 Treatments



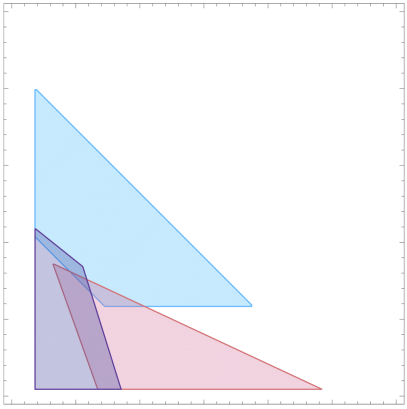
(a) Hypothetical $\Pi_{\mathbf{T}|1}$

(Stylized) Example with 3 Treatments



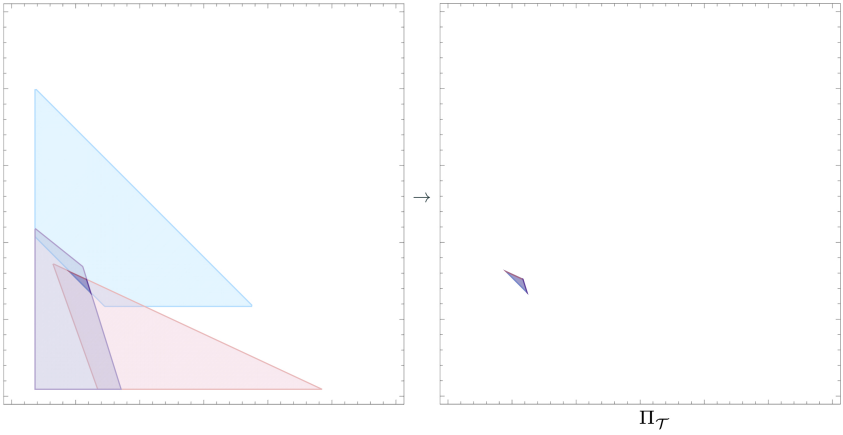
(a) Hypothetical $\Pi_{\mathbf{T}|1}, \Pi_{\mathbf{T}|2}$

(Stylized) Example with 3 Treatments



(a) Hypothetical $\Pi_{\mathbf{T}|1}, \Pi_{\mathbf{T}|2}, \Pi_{\mathbf{T}|3}$

(Stylized) Example with 3 Treatments



(a) Hypothetical $\Pi_{\mathbf{T}} := \bigcap_{d \in \mathcal{D}} \Pi_{\mathbf{T}|d}$

Identified Set for π

Since $\int_{\mathcal{Y}} \boldsymbol{q}_d(y)\, dy = \boldsymbol{\pi}$, and $\boldsymbol{q}_d$ satisfies ($\star$), we must have

$$\boldsymbol{\pi} \in \left\{ \int_{\mathcal{Y}} \boldsymbol{x}(y)\, dy \;\middle|\; \begin{array}{l} \text{integrable } \boldsymbol{x} : \mathcal{Y} \to \mathbb{R}^{\mathcal{T}}_{\geq 0} \\ \mathbf{R}_d \boldsymbol{x}(y) = \boldsymbol{p}_d(y)\,,\ \forall y \in \mathcal{Y} \end{array} \right\} =: \Pi_d$$

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$\rightarrow$ This must hold for all d, so $\boldsymbol{\pi} \in \bigcap_{d \in \mathcal{D}} \Pi_d =: \Pi_{\mathcal{T}}$

$\rightarrow$ To satisfy $\mathbf{S}$, also need $\boldsymbol{\pi} \in \mathcal{R}_{\boldsymbol{\pi}}$

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Theorem 2 (Sharp Testable Implications of Selection Models)

For any treatment selection model $\mathbf{S}$,

$$\Pi_{\mathbf{S}} := \left\{ \boldsymbol{\pi}_Q \,\middle|\, Q \in \mathcal{Q}_{\mathbf{S}} \right\} = \mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S}) \cap \Pi_{\mathcal{T}(\mathbf{S})}$$

In particular, $\mathcal{Q}_{\mathbf{S}} \neq \emptyset$ if and only if $\mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S}) \cap \Pi_{\mathcal{T}(\mathbf{S})} \neq \emptyset$.

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In particular, $\mathcal{Q}_{\mathbf{S}} \neq \emptyset$ if and only if $\mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S}) \cap \Pi_{\mathcal{T}(\mathbf{S})} \neq \emptyset$.

$\rightarrow$ **Goal:** Tractable representation of $\mathcal{R}_{\boldsymbol{\pi}} \cap \Pi_{\mathcal{T}}$

Candidates for π: Support Function

Every non-empty closed convex set $C \subseteq \mathbb{R}^n$ is uniquely determined by its ***support function*** $\boldsymbol{h}_C : \mathbb{R}^n \to \mathbb{R}$, defined as

$$\boldsymbol{h}_C(\boldsymbol{c}) := \sup_{\boldsymbol{x} \in C} \boldsymbol{c}^\top \boldsymbol{x} \ .$$

Define the shorthand $\boldsymbol{h}_d(\boldsymbol{c}; \boldsymbol{p}_d(y)) := \sup_{\boldsymbol{x} \geq 0 : \mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)} \boldsymbol{c}^\top \boldsymbol{x}$.

Candidates for π: Support Function

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Define the shorthand $h_d(\boldsymbol{c}; \boldsymbol{p}_d(y)) := \sup_{\boldsymbol{x} \geq 0 : \mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)} \boldsymbol{c}^\top \boldsymbol{x}$.

Lemma 1

Suppose $\mathcal{T}$ and P are such that for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)$. Then, Π_d is a non-empty convex set such that, for all $\boldsymbol{c} \in \mathbb{R}^{\mathcal{T}}$

$$h_{\Pi_d}(\boldsymbol{c}) = \int_{\mathcal{Y}} h_d(\boldsymbol{c}; \boldsymbol{p}_d(y)) \ dy \ .$$

Further, Π_d is a convex polytope.

Candidates for π: Moment Representation

Lemma 2

Suppose for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)$. Then, for $\boldsymbol{c} : \mathcal{Y} \to \mathbb{R}^{\mathcal{T}}$ with $\boldsymbol{c}_{(\boldsymbol{\tau})}(\cdot) = 0$ for all "d-never-takers", there exists $\boldsymbol{m}_d(\boldsymbol{c}(\cdot), \boldsymbol{p}_d(\cdot))$ such that, for all $y \in \mathcal{Y}$,

$$\boldsymbol{m}_d(\boldsymbol{c}(y); \boldsymbol{p}_d(y)) \in \underset{\substack{\boldsymbol{\lambda}: \|\boldsymbol{\lambda}\|_\infty \leq M \|\boldsymbol{c}(y)\|_\infty, \\ \mathbf{R}_d^\top \boldsymbol{\lambda} \geq \boldsymbol{c}(y)}}{\arg\min} \boldsymbol{\lambda}^\top \boldsymbol{p}_d(y) \,,$$

with M a constant (does not depend on P or $\boldsymbol{c}$, only $\mathbf{R}_d$)

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with M a constant (does not depend on P or $\boldsymbol{c}$, only $\mathbf{R}_d$), and

$$\begin{aligned} &\int_{\mathcal{Y}} \boldsymbol{h}_d(\boldsymbol{c}(y), \boldsymbol{p}_d(y)) \, dy \\ &\quad = \sum_{z \in \mathcal{Z}} \mathbb{E}\left[\boldsymbol{m}_{d,(z)}(\boldsymbol{c}(Y); \boldsymbol{p}_d(Y)) \mathbb{1}\{D = d\} \,\middle|\, Z = z\right] \\ &\quad = \mathbb{E}\left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D = d, Z = z\}}{\mathrm{P}(Z = z)} \boldsymbol{m}_{d,(z)}(\boldsymbol{c}(Y); \boldsymbol{p}_d(Y))\right]. \end{aligned}$$

Candidates for π: Moment Representation

For any conformable matrix $\boldsymbol{B}$, with rows $\{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_k\}$, define shorthands

$$\int_{\mathcal{Y}} \boldsymbol{h}_d\left(\boldsymbol{B}; \boldsymbol{p}_d(y)\right) dy := \begin{bmatrix} \int_{\mathcal{Y}} \boldsymbol{h}_d\left(\boldsymbol{b}_1; \boldsymbol{p}_d(y)\right) dy \\ \vdots \\ \int_{\mathcal{Y}} \boldsymbol{h}_d\left(\boldsymbol{b}_k; \boldsymbol{p}_d(y)\right) dy \end{bmatrix},$$

$$\mathbb{E}\left[\boldsymbol{m}_d\left(\boldsymbol{B}; Y, D, Z\right)\right] := \begin{bmatrix} \mathbb{E}\left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D=d, Z=z\}}{\mathrm{P}(Z=z)} \boldsymbol{m}_d\left(\boldsymbol{b}_1; \boldsymbol{p}_d(Y)\right)_{(z)}\right] \\ \vdots \\ \mathbb{E}\left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D=d, Z=z\}}{\mathrm{P}(Z=z)} \boldsymbol{m}_d\left(\boldsymbol{b}_k; \boldsymbol{p}_d(Y)\right)_{(z)}\right] \end{bmatrix}.$$

Candidates for π: Moment Representation

Corollary 1

There exists a known matrix $\mathbf{A}_d$, depending only on $\mathbf{R}_d$, such that, if for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)$, then

$$\begin{aligned}\Pi_d &= \left\{ \boldsymbol{x} \in \mathbb{R}^{\mathcal{T}} \;\middle|\; \mathbf{A}_d \boldsymbol{x} \leq \int_{\mathcal{Y}} \boldsymbol{h}_d(\mathbf{A}_d; \boldsymbol{p}_d(y))\, dy \right\} \\ &= \left\{ \boldsymbol{x} \in \mathbb{R}^{\mathcal{T}} \;\middle|\; \mathbf{A}_d \boldsymbol{x} \leq \mathbb{E}\left[\boldsymbol{m}_d(\mathbf{A}_d; Y, D, Z)\right] \right\} .\end{aligned}$$

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Define $\mathbf{A}_{\mathcal{T}} := \begin{bmatrix} \mathbf{A}_1 \\ \vdots \\ \mathbf{A}_{N_D} \end{bmatrix}$. If the condition holds for all $d \in \mathcal{D}$, then,

$$\Pi_{\mathcal{T}} = \left\{ \boldsymbol{x} \in \mathbb{R}^{\mathcal{T}} \;\middle|\; \mathbf{A}_{\mathcal{T}} \boldsymbol{x} \leq \begin{bmatrix} \mathbb{E}\left[\boldsymbol{m}_1\left(\mathbf{A}_1; Y, D, Z\right)\right] \\ \vdots \\ \mathbb{E}\left[\boldsymbol{m}_{N_D}\left(\mathbf{A}_{N_D}; Y, D, Z\right)\right] \end{bmatrix} \right\} ,$$

and $\Pi_{\mathcal{T}}$ is a compact convex polytope.

Candidates for π: Moment Representation

Corollary 1

There exists a known matrix $\mathbf{A}_d$, depending only on $\mathbf{R}_d$, such that, if for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)$, then

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and $\Pi_{\mathcal{T}}$ is a compact convex polytope.

$\rightarrow \Pi_{\mathcal{T}} \neq \emptyset$ iff above finite linear system has a solution

Sharp Specification Test for Treatment Selection Models

Define $\mathbb{E}\left[\boldsymbol{m}_{\mathcal{T}}\left(Y,D,Z\right)\right] := \begin{bmatrix} \mathbb{E}[\boldsymbol{m}_1(\mathbf{A}_1;Y,D,Z)] \\ \vdots \\ \mathbb{E}\left[\boldsymbol{m}_{N_D}\left(\mathbf{A}_{N_D};Y,D,Z\right)\right] \end{bmatrix}$.

Theorem 3 (Specification Test for Convex $\mathcal{R}_{\boldsymbol{\pi}}$)

Suppose $\mathcal{R}_{\boldsymbol{\pi}}\left(\mathbf{S}\right)$ is a closed convex set and, for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d\boldsymbol{x} = \boldsymbol{p}_d\left(y\right)$. Then, for all $\boldsymbol{c} \in \mathbb{R}^{\mathcal{T}}$,

$$\begin{aligned} \boldsymbol{h}_{\Pi_{\mathbf{S}}}\left(\boldsymbol{c}\right) &= \inf_{\boldsymbol{u}\in\mathbb{R}^{\mathcal{T}}} \left\{\boldsymbol{h}_{\mathcal{R}_{\boldsymbol{\pi}}}\left(\boldsymbol{c}-\boldsymbol{u}\right) + \boldsymbol{h}_{\Pi_{\mathcal{T}}}\left(\boldsymbol{u}\right)\right\} \\ &= \inf_{\boldsymbol{\lambda}\geq 0} \left\{\boldsymbol{h}_{\mathcal{R}_{\boldsymbol{\pi}}}\left(\boldsymbol{c}-\mathbf{A}_{\mathcal{T}}^{\top}\boldsymbol{\lambda}\right) + \boldsymbol{\lambda}^{\top}\mathbb{E}\left[\boldsymbol{m}_{\mathcal{T}}\left(Y,D,Z\right)\right]\right\} . \end{aligned}$$

Further,

$$\mathcal{Q}_{\mathbf{S}} \neq \emptyset \iff \inf_{\boldsymbol{\lambda}\geq 0:\|\mathbf{A}_{\mathcal{T}}^{\top}\boldsymbol{\lambda}\|_1=1} \left\{\boldsymbol{h}_{\mathcal{R}_{\boldsymbol{\pi}}}\left(-\mathbf{A}_{\mathcal{T}}^{\top}\boldsymbol{\lambda}\right) + \boldsymbol{\lambda}^{\top}\mathbb{E}\left[\boldsymbol{m}_{\mathcal{T}}\left(Y,D,Z\right)\right]\right\} \geq 0.$$

Sharp Specification Test for Treatment Selection Models

When restrictions implied by $\mathcal{R}_{\boldsymbol{\pi}}$ are linear, $\mathcal{R}_{\boldsymbol{\pi}}$ is a convex polytope:

$$\mathcal{R}_{\boldsymbol{\pi}} = \left\{ \boldsymbol{\pi} \in [0,1]^{\mathcal{T}} \;\middle|\; \mathbf{A}_{\mathcal{R}}\, \boldsymbol{\pi} \leq \mathbf{a}_{\mathcal{R}} \right\}.$$

Sharp Specification Test for Treatment Selection Models

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Let $W := (Y, D, Z)$, $\mathbf{A}_{\mathbf{S}} := \begin{bmatrix} \mathbf{A}_{\mathcal{T}} \\ \mathbf{A}_{\mathcal{R}} \end{bmatrix}$ and $\mathbb{E}\left[\boldsymbol{m}_{\mathbf{S}}(W)\right] := \begin{bmatrix} \mathbb{E}[\boldsymbol{m}_{\mathcal{T}}(W)] \\ \mathbf{a}_{\mathcal{R}} \end{bmatrix}$.

Theorem 4 (Specification Test for Linear $\mathcal{R}_{\boldsymbol{\pi}}$)

Suppose $\mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S})$ is a closed convex polytope as above and, for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)$. Then, for all $\boldsymbol{c} \in \mathbb{R}^{\mathcal{T}}$,

$$\boldsymbol{h}_{\Pi_{\mathbf{S}}}(\boldsymbol{c}) = \inf_{\boldsymbol{\lambda} \geq 0 : \mathbf{A}_{\mathbf{S}}^{\top} \boldsymbol{\lambda} = \boldsymbol{c}} \boldsymbol{\lambda}^{\top} \mathbb{E}\left[\boldsymbol{m}_{\mathbf{S}}(W)\right].$$

Further, $\quad \mathcal{Q}_{\mathbf{S}} \neq \emptyset \iff \inf_{\substack{\boldsymbol{\lambda} \geq 0 : \|\boldsymbol{\lambda}\|_1 = 1, \\ \mathbf{A}_{\mathbf{S}}^{\top} \boldsymbol{\lambda} = \mathbf{0}}} \boldsymbol{\lambda}^{\top} \mathbb{E}\left[\boldsymbol{m}_{\mathbf{S}}(W)\right] \geq 0.$

Sharp Specification Test, II

When matrices $\mathbf{A}_d$ not available, can use following alt. formulation.

Theorem 4B

Suppose $\mathcal{R}_{\boldsymbol{\pi}}(\mathbf{S})$ is a closed convex polytope and, for all $y \in \mathcal{Y}$, there exists $\boldsymbol{x} \geq 0$ with $\mathbf{R}_d \boldsymbol{x} = \boldsymbol{p}_d(y)$. Then, for all $\boldsymbol{c} \in \mathbb{R}^{\mathcal{T}}$,

$$h_{\Pi_{\mathbf{S}}}(\boldsymbol{c}) = \inf_{\substack{(\boldsymbol{w}_d : d): \\ \boldsymbol{w}_d \in \mathbb{R}^{\mathcal{T}}}} \left\{ \mathbf{a}_{\mathcal{R}}^{\top} \left(\boldsymbol{c} - \sum_d \boldsymbol{w}_d \right) + \sum_{d \in \mathcal{D}} \mathbb{E}\left[\boldsymbol{m}_d(\boldsymbol{w}_d; W) \right] \right\}$$

Further,

$$\mathcal{Q}_{\mathbf{S}} \neq \emptyset \iff \inf_{\substack{(\boldsymbol{w}_d : d): \boldsymbol{w}_d \in \mathbb{R}^{\mathcal{T}}, \\ \|(\boldsymbol{w}_d : d)\|_\infty \leq 1}} \left\{ \mathbf{a}_{\mathcal{R}}^{\top} \left(\sum_d \boldsymbol{w}_d \right) + \sum_{d \in \mathcal{D}} \mathbb{E}\left[\boldsymbol{m}_d(\boldsymbol{w}_d; W) \right] \right\} \geq 0.$$

Inclusion Test for Mean Outcomes: Moment Representation

When $\mathcal{R}_{\boldsymbol{\pi}} = \{\boldsymbol{\pi} \mid \mathbf{A}_{\mathcal{R}}\ \boldsymbol{\pi} \leq \mathbf{a}_{\mathcal{R}}\}$, with $W := (Y, D, Z)$:

$$\Theta_{\mathbf{S}} = \left\{ \gamma \in \Theta_0 \;\middle|\; \inf_{\substack{(t,\boldsymbol{\lambda}):\boldsymbol{\lambda}\geq 0,\\ \|(t,\boldsymbol{\lambda})\|_\infty \leq 1}} \underbrace{\left\{ \begin{array}{r} \mathbb{E}\left[\boldsymbol{m}_d\left(t\left(\kappa(Y)-\gamma\right)\boldsymbol{c} - \mathbf{A}_{\mathbf{S}}^\top \boldsymbol{\lambda}; W\right)\right] \\ + \boldsymbol{\lambda}^\top \mathbb{E}\left[\boldsymbol{m}_{\mathbf{S}}\left(W\right)\right] \end{array} \right\}}_{=:\ \Psi_\gamma(t,\boldsymbol{\lambda})} \geq 0 \right\}$$

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Let $\underline{\Psi}_\gamma$ be infimum of unconditional moment inequalities problem above.

$\rightarrow$ Can construct confidence region for θ by inverting, over γ, test of

$$H_0 : \underline{\Psi}_\gamma \geq 0 \quad \text{vs.} \quad H_1 : \underline{\Psi}_\gamma < 0$$

$\rightarrow$ Ψ_γ convex in $(t, \boldsymbol{\lambda}) \implies$ finite-dimensional convex program

$\rightarrow$ $\boldsymbol{m}_d$ functions depend on P and must be estimated

Computation of m_d and h_d: Setup

Given vector (or matrix) $\boldsymbol{x}$ and a subset B of coordinates of $\boldsymbol{x}$, let:

- $\boldsymbol{x}_B$ be the subvector of $\boldsymbol{x}$ corresponding to the coordinates in B
- $\boldsymbol{x}_{B^c}$ be the subvector consisting of the remaining coordinates

"Basic solution" to (primal) LP $\max_{\boldsymbol{x} \geq 0 : \mathbf{R}_d \boldsymbol{x} = \boldsymbol{b}} \boldsymbol{c}^\top \boldsymbol{x}$ is of the form

$$\boldsymbol{x}_B = \mathbf{R}_{d,B}^{-1} \boldsymbol{b} \text{ and } \boldsymbol{x}_{B^c} = \mathbf{0} ,$$

where B is a subset of columns of $\mathbf{R}_d$ such that $\mathbf{R}_{d,B}$ is invertible.

Basic solution optimal iff both of the following hold,

1. It is primal-feasible: $\mathbf{R}_{d,B}^{-1} \boldsymbol{b} \geq \mathbf{0}$,
2. It is dual-feasible: $\boldsymbol{c}_B^\top \mathbf{R}_{d,B}^{-1} \mathbf{R}_{d,B^c} - \boldsymbol{c}_{B^c}^\top \geq \mathbf{0}$.

$\rightarrow$ **Key:** Solve LP at one $y' \in \mathcal{Y}$ $\implies$ get sol'ns for y-values in *region* of $\mathcal{Y}$

Let $P_d(B) := \left\{ \boldsymbol{b} \,|\, \mathbf{R}_{d,B}^{-1} \boldsymbol{b} \geq \mathbf{0} \right\}$, $D_d(B) := \left\{ \boldsymbol{c} \,|\, \boldsymbol{c}_B^\top \mathbf{R}_{d,B}^{-1} \mathbf{R}_{d,B^c} - \boldsymbol{c}_{B^c}^\top \geq \mathbf{0} \right\}$.

Computation of m_d and h_d: Algorithm

Algorithm 1

Input Functions $\boldsymbol{c} : \mathcal{Y} \to \mathbb{R}^{\mathcal{T}}$ and $\boldsymbol{p}_d : \mathcal{Y} \to \mathbb{R}^{\mathcal{Z}}$,
Solver $\mathrm{LP}_d(\boldsymbol{c}, \boldsymbol{b})$, returns an optimal basis for $\max_{\boldsymbol{x} \geq 0 : \mathbf{R}_d \boldsymbol{x} = \boldsymbol{b}} \boldsymbol{c}^\top \boldsymbol{x}$.

Output Functions $\boldsymbol{h}_d : \mathcal{Y} \to \mathbb{R}$ and $\boldsymbol{m}_d : \mathcal{Y} \to \mathbb{R}^{\mathcal{Z}}$.

1: Set $\mathcal{Y}^\dagger := \emptyset$ and $\mathcal{G} := \emptyset$.

2: **While** $\mathcal{Y}^\dagger \neq \mathcal{Y}$:

3: Choose arbitrary $\tilde{y} \in \mathcal{Y} \setminus \mathcal{Y}^\dagger$ and update $\mathcal{G} \leftarrow \mathcal{G} \cup \{\tilde{y}\}$,

4: Set $B(\tilde{y}) = \mathrm{LP}_d(\boldsymbol{c}(y), \boldsymbol{p}_d(y))$,

5: Set $O(\tilde{y}) := \{y \in \mathcal{Y} \mid \boldsymbol{p}_d(y) \in P_d(B(\tilde{y})),\ \boldsymbol{c}(y) \in D_d(B(\tilde{y}))\} \setminus \mathcal{Y}^\dagger$,

6: Update $\mathcal{Y}^\dagger \leftarrow \mathcal{Y}^\dagger \sqcup O(\tilde{y})$.

7: **Return** $\boldsymbol{h}_d(y) = \sum_{\tilde{y} \in \mathcal{G}} \mathbb{1}\{y \in O(\tilde{y})\}\, \boldsymbol{c}(y)^\top \mathbf{R}_{d,B(\tilde{y})}^{-1} \boldsymbol{p}_d(y)$,
$\boldsymbol{m}_d(y) = \sum_{\tilde{y} \in \mathcal{G}} \mathbb{1}\{y \in O(\tilde{y})\}\, \mathbf{R}_{d,B(\tilde{y})}^{-\top} \boldsymbol{c}(y)$.

Example 1: Head Start Impact Study

Let D denote preschool choice. Families choose between no preschool (n), competing preschools (c), and Head Start (h) for their child.

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The Head Start Impact Study randomized applicants into two groups:

$Z = 0$: Not offered admission to Head Start

$Z = 1$: Offered admission to Head Start.

$\rightarrow$ Therefore, $D \in \{n, c, h\}$ and $Z \in \{0, 1\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = (D_0, D_1)$.

Example 1 (Head Start): Partial Identification

Assume that the following assumption holds for everyone

- either $D_1 = D_0$ or $\mathbb{1}\{D_1 = h\} > \mathbb{1}\{D_0 = h\}$

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The following weighted average can be (point) identified

$$\omega_{n\to h} E\left[Y_h - Y_n \mid D_0 = n, D_1 = h\right] + \omega_{c\to h} E\left[Y_h - Y_c \mid D_0 = c, D_1 = h\right] ,$$

but the two component effects are only partially identified.

Example 1 (Head Start): "h"-response-type

Consider the effect $E\left[Y_h - Y_c \mid \boldsymbol{T} = (ch)\right]$

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$\rightarrow$ Therefore, it suffices to find the bounds for

$$P\left[\boldsymbol{T} = (ch)\right] E\left[Y_h \mid \boldsymbol{T} = (ch)\right]$$

Aggregate $Y_h | \boldsymbol{T} \in \{(ch), (nh)\}$ is point-identified.

$\rightarrow$ Trim the identified density $P\left[\boldsymbol{T} \in \{(ch), (nh)\}\right] f_{Y_h | \boldsymbol{T} \in \{(ch),(nh)\}}(y)$

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There are two types such that $D_0 \neq h$ and $D_1 = h$:

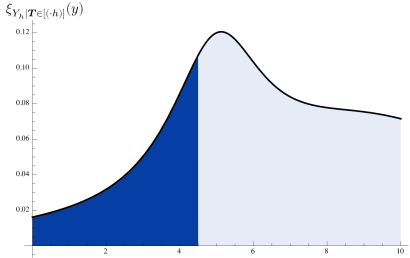
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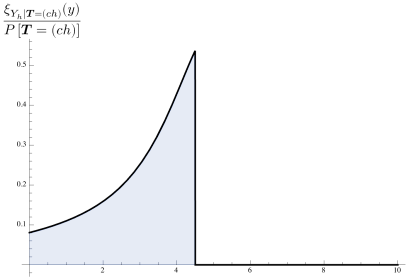
$\rightarrow$ Trim the identified density $P\left[\boldsymbol{T} \in \{(ch), (nh)\}\right] f_{Y_h | \boldsymbol{T} \in \{(ch),(nh)\}}(y)$

Example 1 (Head Start): Sharp Lower Bound

Suppose $P[\boldsymbol{T} = (ch)] = 0.2$.



(a) Dark blue region has area 0.2.



(b) Density $\frac{q_{Y_h|\boldsymbol{T}=ch)}}{P[\boldsymbol{T}=(ch)]} = f_{Y_h|\boldsymbol{T}=(ch)}$ implied by sharp lower bound

Figure 5: (Simulated) Construction of sharp lower bound for $E[Y_h|\boldsymbol{T} = (ch)]$

Example: Moving to Opportunity (MTO) Experiment

Let D denote neighbourhood choice. Here, individuals choose from three different types of neighbourhoods: "high" poverty (H), "medium" poverty (M), and "low" poverty (L).

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Let $Z = (Z_M, Z_L)$ be the randomly assigned instrument. Individuals living in high poverty neighbourhoods were randomly offered rent vouchers:

$(Z_M, Z_L) = (0, 0)$: No vouchers

$(Z_M, Z_L) = (0, 1)$: Vouchers for L neighbourhoods

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$(Z_M, Z_L) = (1,1)$: Vouchers for M and L neighbourhoods

$\rightarrow$ Therefore, $D \in \{H, M, L\}$ and $Z \in \{(0,0), (0,1), (1,1)\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = \left(D_{(0,0)}, D_{(0,1)}, D_{(1,1)}\right)$.

Example: Moving to Opportunity (MTO) Experiment

Suppose we impose "Unordered Partial Monotonicity" (Mountjoy, 2022):

- $\rightarrow$ Z_M incentivizes $D = M$ and Z_L incentivizes $D = L$
- $\rightarrow$ Changes in Z don't effect choices among irrelevant alternatives (e.g., $Z : (0,0) \rightarrow (0,1)$ cannot cause $D : H \rightarrow M$)

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Shifting $Z : (1,1) \rightarrow (0,1)$ causes two types of moves:

- Individuals with $\boldsymbol{T} \in \{(H,L,M), (M,L,M), (L,L,M)\}$: $M \rightarrow L$
- Individuals with $\boldsymbol{T} = (H,H,M)$: $M \rightarrow H$

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The following weighted average can be (point) identified

$$\omega_{M\rightarrow L} E\left[Y_L - Y_M \mid D_{(0,1)} = L, D_{(1,1)} = M\right] + \omega_{M\rightarrow H} E\left[Y_H - Y_M \mid D_{(0,1)} = H, D_{(1,1)} = M\right],$$

but the two component effects are only partially identified.

Example 2 (MTO): "M"-response-type

Recall: $D \in \{H, M, L\}$ and $Z = (Z_M, Z_L) \in \{(0,0), (0,1), (1,1)\}$.
Response-type: $\boldsymbol{T} = \left(D_{(0,0)} D_{(0,1)} D_{(1,1)}\right)$.

Consider the effect $E\left[Y_H - Y_M \mid \boldsymbol{T} = (HHM)\right]$

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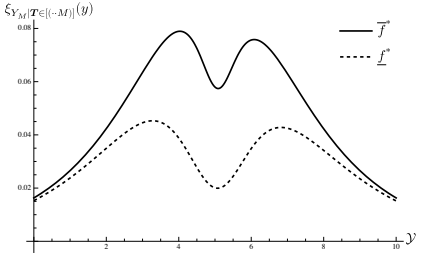
There are three types such that $D_{(0,0)} \neq M, D_{(0,1)} \neq M$ and $D_{(1,1)} = M$:

$$\{(HHM), (HLM), (LLM)\}$$

Dist. of Y_M for these types is not identified, but can be bounded

Example 2 (MTO): Non-Sharpness of Simple Bounds

Suppose the sharp upper bound on $P[\boldsymbol{T} \in [(\cdot\,\cdot M)]]$ is 0.36, and $P[\boldsymbol{T} = (HHM)] = 0.28$.



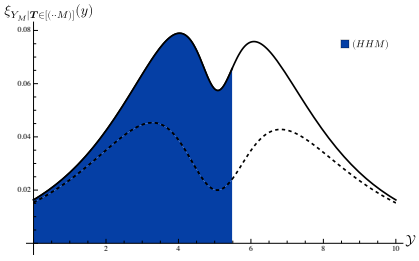
Pointwise bounds on $\boldsymbol{q}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]}$

Figure 6: (Simulated) Construction of lower bounds for $E[Y_M|\boldsymbol{T} = (HHM)]$

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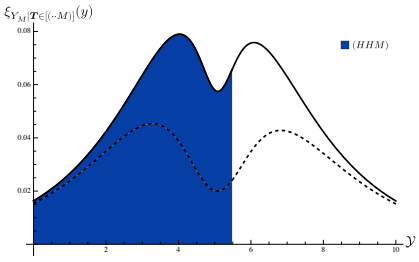
dark blue area $= 0.28$

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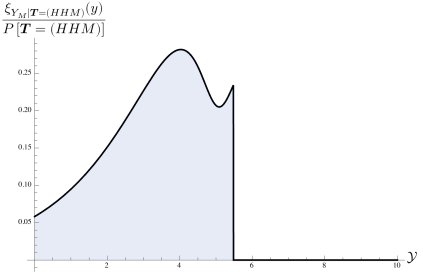
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Construction of Simple Lower Bound



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Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Simple Lower Bound



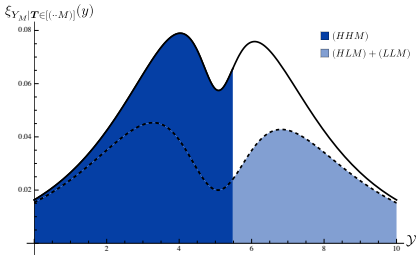
This density implies the simple lower bound of 3.289

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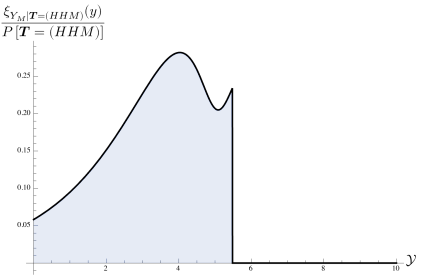
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Construction of Simple Lower Bound



dark blue area $= 0.28$
dark blue area + light blue area $= \mathbf{0.423}$

Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Simple Lower Bound



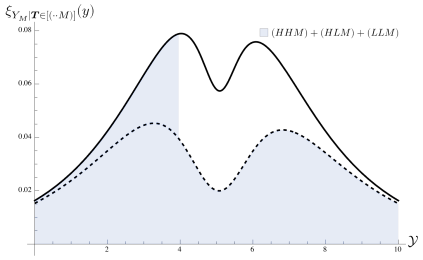
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Construction of Sharp Lower Bound



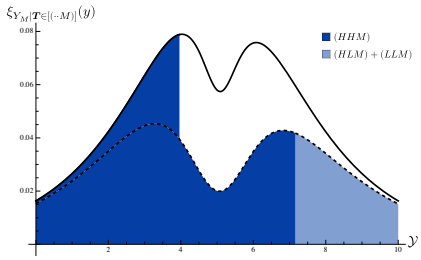
shaded area $= \mathbf{0.36}$

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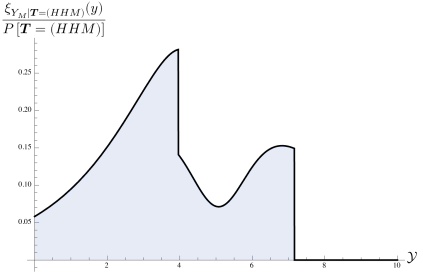
Suppose the sharp upper bound on $P[\boldsymbol{T} \in [(\cdot\cdot M)]]$ is 0.36, and $P[\boldsymbol{T} = (HHM)] = 0.28$.

Construction of Sharp Lower Bound



dark blue area $= 0.28$
dark blue area + light blue area $= \mathbf{0.36}$

Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Sharp Lower Bound



This density implies the sharp lower bound of 3.665

Figure 6: (Simulated) Construction of lower bounds for $E[Y_M|\boldsymbol{T} = (HHM)]$

Example 3: Multi-dimensional Treatment

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Some individuals are randomly offered incentives to participate in treatment. Let $Z = (Z_P, Z_S)$ where

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Suppose individuals are

- only offered supplementary incentive if offered primary incentive
- only allowed supplementary treatment if they take primary treatment

Example 3: Multi-dimensional Treatment

Consider a setting with a primary treatment, $P \in \{0,1\}$, and supplementary treatment, $S \in \{0,1\}$, and define $D = (P,S)$.

Some individuals are randomly offered incentives to participate in treatment. Let $Z = (Z_P, Z_S)$ where

- $Z_P \in \{0,1\}$ represents an incentive to participate in P
- $Z_S \in \{0,1\}$ represents an additional incentive to participate in S

Suppose individuals are

- only offered supplementary incentive if offered primary incentive
- only allowed supplementary treatment if they take primary treatment

$\rightarrow$ Therefore, $D \in \{(0,0),(1,0),(1,1)\}$ and $Z \in \{(0,0),(1,0),(1,1)\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = \left(D_{(0,0)}, D_{(1,0)}, D_{(1,1)}\right)$.

Multi-dimensional Treatment Example

Assume that the following holds for all individuals:

1. $Z_P : 0 \to 1$ (weakly) increases $\mathbb{1}\{D \in \{(1,0),(1,1)\}\}$ and $\mathbb{1}\{S = 1\}$,
2. $Z_S : 0 \to 1$ (weakly) increases $\mathbb{1}\{D = (1,1)\}$ and $\mathbb{1}\{P = 1\}$.

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For example, consider the decomposition

$$\underbrace{E\left[Y_{(1,1)} - Y_{(0,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Total Effect}}$$
$$= \underbrace{E\left[Y_{(1,1)} - Y_{(1,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Supplementary Effect}}$$
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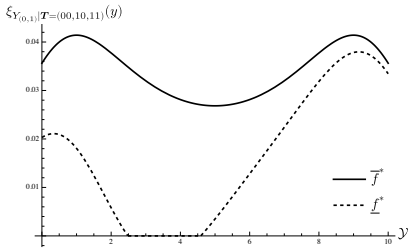
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$\to$ **All components are only partially identified!**

Multi-dim. Treatment example: Tighter Bounds due to Pointwise Dependence

Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.



Pointwise bounds on $\boldsymbol{q}_{\boldsymbol{Y}_M|\boldsymbol{T}=(00,10,11)}$

Figure 7: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

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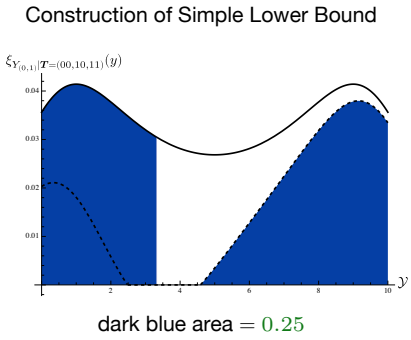
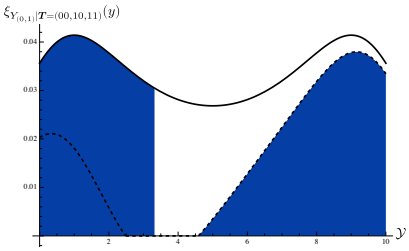


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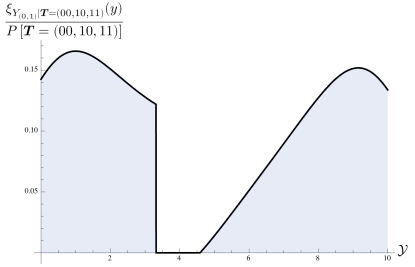
Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.

Construction of Simple Lower Bound



dark blue area $= 0.25$

Density $f_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ Implied by Simple Lower Bound



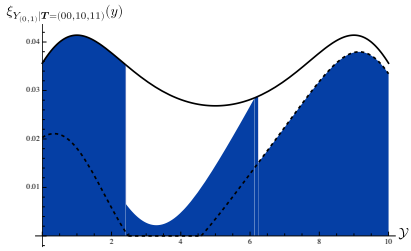
This density implies the simple lower bound of 4.865

Figure 7: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

Multi-dim. Treatment example: Tighter Bounds due to Pointwise Dependence

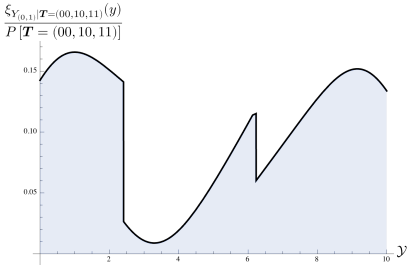
Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.

Construction of Sharp Lower Bound



dark blue area $= 0.25$

Density $f_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ Implied by Sharp Lower Bound



This density implies the sharp lower bound of 5.106

Figure 7: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

Estimating the Bounds

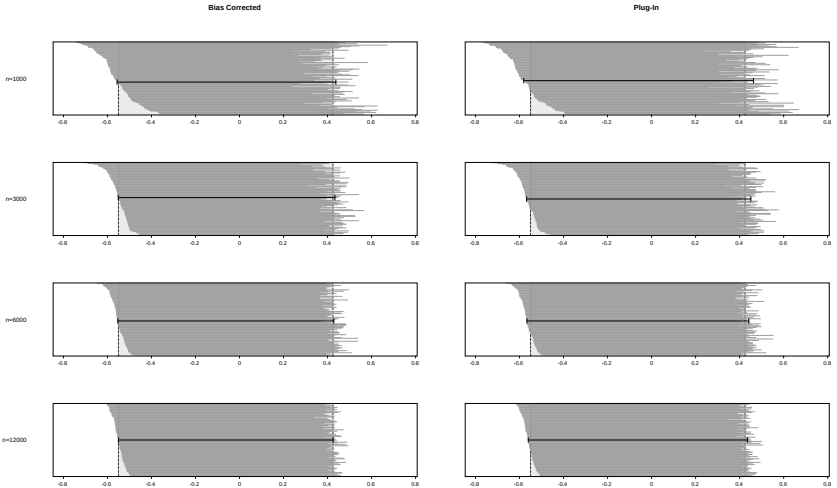
Generally, sharp upper bound $\bar{\theta}(P)$ defined by $\overline{\Psi}\left(\bar{\theta}(P); P\right) = 0$

Where $\gamma \mapsto \overline{\Psi}(\gamma; P)$ is a decreasing function of the form

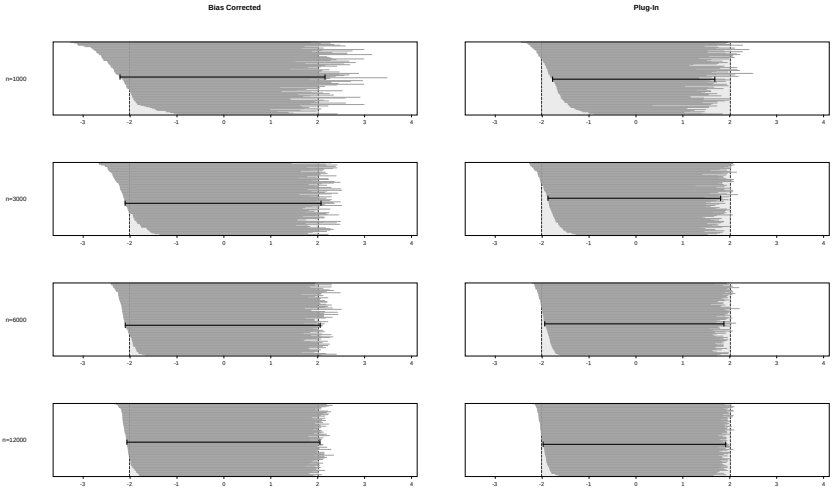
$$\overline{\Psi}(\gamma; P) := \inf_{\boldsymbol{u} \in <\text{known subset of } \mathbb{R}^p>} \sum_{z \in \mathcal{Z}} \mathbb{E}_P \left[\boldsymbol{m}_\theta^P(\gamma, \boldsymbol{u}; Y, D, Z) \,\middle|\, Z = z \right] .$$

- → In practice, don't know P, so can "plug-in" estimate $\tilde{P}_n$
- → Infimum of estimated objective $\implies \mathbb{E}_P\left[\overline{\Psi}\left(\gamma; \tilde{P}_n\right)\right] < \overline{\Psi}(\gamma; P)$
- → $\overline{\Psi}\left(\bar{\theta}(P); P\right) = 0 = \overline{\Psi}\left(\bar{\theta}\left(\tilde{P}_n\right); \tilde{P}\right) < \overline{\Psi}\left(\bar{\theta}\left(\tilde{P}_n\right); P\right)$
- → $\implies \bar{\theta}\left(\tilde{P}_n\right) < \bar{\theta}(P)$, since $\overline{\Psi}$ decreasing
- → **In paper:** bias-corrected estimator!

Estimator Performance, Regular Example (DGP from BHMSSV)

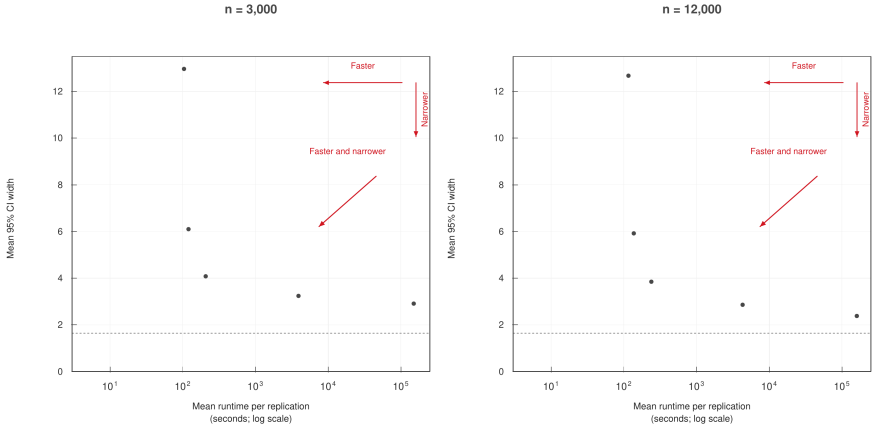


Estimator Performance, Irregular Example



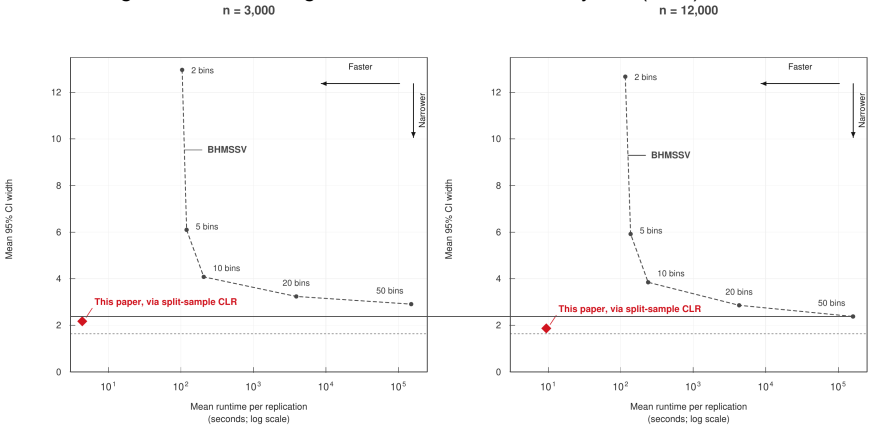
Inference: Comparison to Discretized Methods

DGP from Figure 1 of Bai, Huang, Moon, Santos, Shaikh, & Vytlacil (2026)

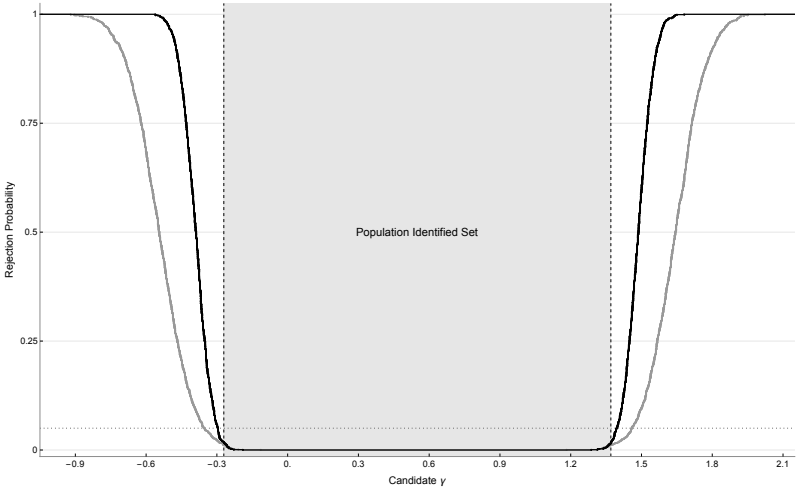


Inference: Comparison to Discretized Methods

DGP from Figure 1 of Bai, Huang, Moon, Santos, Shaikh, & Vytlacil (2026)



Inference: Power Curves



Grey = 3k, Black = 12k; 1300 Samples.

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